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Quantifying the Impact of Resource Redundancy on Smart City System Dependability: A Model-Driven Approach

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Abstract Effective quality management plays a pivotal role in ensuring the smooth operation of smart city systems, which have significant implications for safety, accessibility, affordability, and maintainability. Dependability of autonomous systems is of utmost importance, as achieving satisfactory levels of availability and reliability poses considerable challenges. Smart cities are characterized by interconnected sub-architectures, encompassing vehicle monitoring, sidewalk monitoring, and building monitoring, all of which need to function efficiently. Analytical models such as Petri nets, Markov chains, and fault trees are well-suited for evaluating complex scenarios in the context of smart cities. This paper presents analytical models that utilize fault tree and Markov chain techniques to assess the availability and reliability of smart city monitoring systems. The model is divided into shared and non-shared components, with non-shared components being specific to certain contextual applications, while shared components, such as data processing and electrical power, are essential for all smart city monitoring and management systems. The study underscores the ease with which the fault tree model can enhance availability by modifying failure requirements and resources. Case studies provide concrete examples of how availability improved from 95.3% to 99.8% by varying a configuration known as "KooN" in multiple components. This paper takes a comprehensive approach to evaluating the dependability of smart city architectures and contributes advancements, such as hierarchical modeling, sequential sensitivity analysis, and the "KooN" analytic method. These contributions expand the

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existing knowledge and methodologies in smart city dependability analysis. Moreover, this work aims to serve as a practical tool to assist smart city managers in optimizing their proposals. All modeling aspects and parameters are detailed thoroughly to enable effective implementation of the proposed approach by anyone using it.

Keywords Quality management, smart city, dependability, fault tree, Markov chain, resource redundancy

1 Introduction

Smart cities have emerged as a prominent initiative by governments worldwide to make cities more navigable and livable, as evidenced by numerous ongoing projects in the public and private sectors [14, 17]. Given the projected growth of the world population by over 10% in the next 30 years [50], quality management is of utmost importance in the development and management of smart cities. Countries around the globe are striving to equip their cities to effectively handle such growth [2]. While cities are hubs of human and economic activity, they also present a plethora of challenges that can become increasingly difficult to solve as they grow in size and complexity. Disparities in cities, particularly in more developed areas, can have negative consequences that may outweigh their positive aspects if not addressed effectively. Urban regions must manage their growth while promoting economic competitiveness, social cohesion, and sustainability.

The concept of a smart city encompasses a local entity, such as a city, region, or small locality, that adopts a holistic approach in employing information technologies with real-time analysis to promote sustainable economic development [28]. The implementation of a smart city entails addressing a myriad of challenges in diverse areas, including healthcare, education, power, transportation, waste management, unemployment, and cyber-security [48]. The fundamental premise of a smart city rests on the utilization of user-friendly information and communication technologies specifically designed for urban areas by major enterprises. This subject matter has garnered substantial attention from both academia and practitioners, emerging as a robust research domain.

Numerous cities across the world are actively pursuing the goal of becoming "smarter" by leveraging Information and Communication Technology (ICT) to enhance various aspects of city operations and management [29]. Notably, cities such as Singapore, Helsinki, and Zurich have been acknowledged as some of the world's leading smart cities in the 2020 Smart City Index [18]. These cities are subject to stringent system requirements, given that human lives are often at stake in multiple facets of a smart city. For instance, in the context of smart traffic, it is imperative for all intelligent traffic lights to function optimally with maximum availability and reliability to prevent catastrophic failures in densely populated urban centers.

Dependability pertains to a system's ability to consistently and accurately deliver a service, considering the potential failure of one or more system components that may impede its ability to provide the intended service. As such, reliability is a crucial attribute of dependability, denoting the probability of a system operating without failure over a specified period of time [45]. Another vital aspect of dependability is availability, which quantifies the probability of a system continuing to function despite the occurrence of failures and subsequent repairs [40].

Ensuring the reliability and availability of highly complex systems poses a formidable challenge. As the number of components in a system increases, predicting its behavior becomes more intricate. Smart cities, for instance, encompass intricate computational architectures that comprise multiple sub-architectures, such as smart buildings, vehicular ad hoc networks (VANETs), and smart surveillance. Each of these sub-architectures encompasses its own set of sub-components, which cannot be analyzed in isolation. A system is deemed available only when all of its sub-components are functional. Consequently, for a smart city to be considered reliable and available, the smart building, VANET, and smart surveillance systems must all operate seamlessly. It is noteworthy that conducting real-world experiments to assess the dependability of smart cities, which typically involve real-time systems, may not always be feasible.

Previous research has not comprehensively addressed the crucial aspect of dependability in the context of smart cities, specifically in analyzing multiple sub-architectures concurrently. Most studies have focused on smart buildings, examining automation mechanisms and Internet of Things (IoT) sensor communication [24, 1, 4, 42, 49]. Only one paper has explored the topic of smart surveillance [19]. In this study, Gonçalves et al. [19] developed stochastic Petri net models for a surveillance monitoring system, analyzing several parameters to achieve the highest level of dependability. Furthermore, only four papers have investigated dependability in the context of vehicular ad-hoc networks (VANETs) [39, 15, 41, 5]. For instance, Qi et al. [39] utilized Time Petri nets (TPN) and Synchronized Petri nets (SPNs) to design a traffic control system (ITCS) for intersections, addressing accidents and providing emergency response to prevent further accidents.

Evaluating the dependability of smart cities can be complex and costly, particularly in real-world environments where a diverse range of components and significant infrastructure are required. Additionally, conducting real experiments in actual environments can have negative impacts on local citizens. To overcome these challenges, formal mathematical models such as queue networks, Markov chains, and Petri nets can be utilized at an early design stage or to evaluate complex configurations in an operating system.

Therefore, the novelty of this work stands out due to its unique approach in assessing the dependability of smart city architectures, which combines fault tree models and Markov chains, adopts a hierarchical modeling strategy, employs KooN gates, and introduces a novel method called sequential sensitivity analysis.

This study contributes to the advancement of modeling and analysis of smart city systems in the following ways specifically:

– The utilization of fault tree (FT) and Markov chain modeling to evaluate the dependability of complex smart city architectures is proposed. FT analysis, a well-established and widely-used technique for evaluating system dependability, is employed [23]. Markov chains, with their mathematical properties, are used in conjunction with FT to model component dependencies [7].
– A hierarchical modeling approach is proposed to represent and calculate the availability and reliability of a smart city control system, using standard macro components such as smart buildings, VANETs, and smart surveillance.
– Advanced sensitivity analysis, known as sequential sensitivity analysis (SSA), is conducted to identify the most impactful components in multiple base model updates.
– Case studies are presented to investigate the impact of changing failure requirements and scalability on availability and reliability metrics.

The structure of the remaining sections in this paper is as follows: Section 2 discusses related works. Section 3 presents the background. Section 4 presents the methodology. Section 5 provides an overview of the architecture of the modeled system. Section 6 presents the proposed FT models, their functionality, and key characteristics. Section 7 presents the results of three case studies, serving as a practical guide for system administrators. Finally, Section 8 concludes the paper and highlights potential areas for future research.

2 Related Work

In this section, a comprehensive literature review pertaining to the proposed work is presented, with a specific focus on the state-of-the-art research in the field of smart cities. The selection criteria for related works primarily considered their relevance to smart cities, as well as their utilization of analytical models. In particular, works that specifically addressed architectural issues related to computational aspects, particularly data processing, were prioritized. To further refine the selection, a final filtering process was conducted to exclude articles that explored topics such as urban mobility [10] and delivery logistics [6], which were not directly pertinent to the technological challenges addressed in this work. The initial search string employed on Google Scholar was as follows: "smart AND (city OR cities OR building OR surveillance OR vanet) AND (Markov OR Petri net OR fault tree OR RBD OR queue)".

In the context of the reviewed literature, the majority of the works identified were focused on the theme of smart buildings. Kharchenko et al. [24] presented a control system for smart buildings that incorporates a three-level architecture for the building automation system, encompassing FPGA, communication, and management components. Abate et al. [1] applied quantitative model checking to analyze the Fault Maintenance Tree (FMT) of a Heating, Ventilation, and

Air-Conditioning (HVAC) unit in a smart building. Specifically, they modeled the FMT using continuous-time Markov chains and priced time automata. Araujo et al. [4] proposed a modeling strategy based on hierarchical models using Stochastic Petri Net (SPN) and Reliability Block Diagram (RBD) to evaluate dependability in an IoT system, utilizing a case study of a smart building scenario. Santos et al. [42] proposed a queuing network-based message exchange architecture to evaluate the performance of an intelligent building infrastructure associated with multiple processing layers, including edge and fog computing. They considered a building with multiple floors and rooms, equipped with sensors and edge devices. Victor et al. [49] proposed a modeling approach based on stochastic Petri nets (SPN) for the quantification of performability in domotics architectures. The SPN performability models were developed based on the architecture of a home automation system that incorporates several IoT sensors, enabling the evaluation of trade-offs between performance and availability.

In the context of smart hospitals, three works were analyzed. Nguyen et al. [36] proposed a comprehensive performability SRN model for an edge/fog-based Medical Information System (MIS) to quantify medical data transactions and services in local hospitals or medical centers. The model incorporated different failure modes of fog nodes and their virtual machines (VMs) under the implementation of fail-over mechanisms. Rodrigues et al. [40] employed analytical models to assess the performance and availability of intelligent hospital systems without the need for real equipment investment. They proposed two Stochastic Petri Net models to represent intelligent hospital architectures. Silva et al. [46] proposed an M/M/c/K queuing network model for the performance evaluation of an Internet of Healthcare Things (IoHT) infrastructure in conjunction with a three-layer cloud/fog/edge computing continuum. The model considered the life cycle of medical data from body-attached IoT sensors in the edge layer to local clients through the fog layer and remote clients through the cloud layer.

Only one publication to date has investigated the topic of smart surveillance, as elucidated in the study conducted by Gonçalves et al. [19]. The authors developed stochastic Petri net models to analyze the dependability of a surveillance monitoring system, by manipulating various parameters to obtain the optimal level of reliability and availability. The dependability of the system was comprehensively assessed, with a particular focus on reliability and availability as the main metrics of interest.

Four scholarly papers have delved into the realm of Vehicular Ad-hoc Networks (VANETs). Qi et al. [39] employed Time Petri nets (TPN) and Synchronized Petri nets (SPNs) to devise a traffic control system for intersections, capable of managing accidents and providing emergency response while preventing further accidents. The proposed system incorporated camera surveillance (CSS) and traffic light control (TLCS) components, which were simulated using SPNs and TPN.

Cesario et al. [15] introduced a "local" synchronization approach for VANETs, whereby computation pertaining to a specific region is performed based on information received from a subset of neighboring regions. This approach offers significant benefits, including reduced computation time, real-time model extraction, and improved support for local decisions.

Rodrigues et al. [41] presented a performance evaluation of a cooperative smart traffic light using a stochastic Petri net (SPN) model. The proposed model enabled the calculation of metrics such as mean response time, resource utilization, and the number of discarded requests. Three case studies were presented to illustrate the utility of the model.

In a study by Araujo et al. [5], stochastic Petri nets (SPNs) and reliability block diagrams (RBD) were employed to assess the availability and reliability of a vehicular cloud computing architecture with multiple Roadside Units (RSUs). Two sensitivity analyses were conducted to identify the components of the model with the most significant impact.

In our research article, we present a novel approach for evaluating the availability and reliability of a complex smart city system, which encompasses three interconnected sub-architectures: smart building, smart surveillance, and VANETs. Our models are developed using a combination of fault tree (FT) and Markov chains, utilizing a hierarchical modeling strategy. This approach enables the simplification of macro-components using FT, while also facilitating detailed analysis of sub-components at a granular level. Additionally, we introduce a unique method called sequential sensitivity analysis (SSA), which allows for the optimization of evaluated metrics by identifying the most critical components of the system. To the best of our knowledge, this is the first study to employ such a strategy. Furthermore, our work is distinguished by the utilization of KooN gates, a distinctive feature inherent to fault tree models, to assess the availability of specific resources and explore diverse failure requirements.

Our proposal places a significant emphasis on achieving a high level of modeling granularity in order to effectively encompass three intricate contexts. However, existing literature has commonly adopted the Petri net approach as a prevailing methodology in this domain, as evidenced by the works of Araujo et al. [4], Victor et al. [49], Nguyen et al. [36], Rodrigues et al. [40], Gonccalves et al. [19], and Qi et al. [39].

Araujo et al. [4] have proposed a modeling strategy based on hierarchical models that utilize Stochastic Petri Net (SPN) and Reliability Block Diagram (RBD) techniques to evaluate the dependability of an IoT system, with a specific case study focused on a smart building scenario. Victor et al. [49] have put forth a modeling approach that employs stochastic Petri nets (SPN) to quantify the performability of domotics architectures, including various IoT sensors/devices, in order to evaluate trade-offs between performance and availability of home automation services. Nguyen et al. [36] have presented a comprehensive performability Stochastic Reward Net (SRN) model for an edge/fog-based medical information system, aimed at quantifying the performability of medical data transactions and services in local hospitals or medical centers.

In addition, Rodrigues et al. [40] have utilized analytical models to assess the performance and availability of intelligent hospital systems, without the need for physical equipment, by proposing two Stochastic Petri Net models to represent intelligent hospital architectures. Gonccalves et al. [19] have developed Stochastic Petri Net models for surveillance monitoring systems, where they varied multiple parameters to achieve the highest dependability, with reliability and availability being the primary metrics of interest. Lastly, Qi et al. [39] have employed Time Petri Nets (TPNs) and Synchronized Petri Nets (SPNs) to design a traffic control system (ITCS) for intersections, specifically aimed at dealing with accidents, providing emergency response, and preventing additional accidents from occurring.

Cesario et al. [15] address the challenge of extracting mobility patterns in an urban computing scenario using stochastic Petri nets (SPN). The authors propose and analyze a "local" synchronization approach, wherein computation for a specific region is performed based on information received from a subset of neighboring regions. Rodrigues et al. [41] present a performance evaluation of a cooperative smart traffic light using an SPN model. The proposed model is capable of calculating the mean response time, resource utilization, and the number of discarded requests. Araujo et al. [5] utilize SPNs and reliability block diagrams (RBD) to assess the availability and reliability of a vehicular cloud computing (VCC) architecture with multiple roadside units (RSUs). Two sensitivity analyses were conducted to identify the model components that have the most significant impact.

Another related context to the current topic in smart cities is railway transportation, in which requires high availability, reliability and security. Two papers applied modelling mechanisms in this context, Lin et al. [30] and Liu et al. [31]. Lin et al. [30] proposed an advanced finite-state Markov chain channel model for high-speed railway, which incorporates the impacts of moving speed on the temporal channel statistical characteristic. The closed-form expressions of state transition probabilities between channel states are derived. The accuracy of the proposed FSMC channel model is validated via extensive measurements conducted on the Zhengzhou-Xi'an HSR line in China. Liu et al. [31] have focused on employing the fault tree analysis method combined with quantitative analysis to investigate high-speed railway accidents. Specifically, by establishing a fault tree logic diagram based on a high-speed railway accident, an in-depth fault tree analysis combined with quantitative analysis is given to present a more comprehensive view of the accident.

Table 1 provides a comparison of related works in the field of smart city systems. Our proposed model stands out due to its unique approach in assessing the dependability of smart city architectures, which combines fault tree models and Markov chains, adopts a hierarchical modeling strategy, employs KooN gates, and introduces a novel method called sequential sensitivity analysis.

Table 1: Related Works.

Paper	**Context**	**Model Types**	**Metrics**	**Hierarchical Modeling**	**Sequential Sensitivity Analysis**	**KooN Analisys**
[24]	Smart building	Markov chain	Security and availability	×	×	×
[1]	Smart building	Fault tree and Markov chain	Availability, reliability and cost	✓	×	×
[4]	Smart building	Petri net and RBD	Availability	✓	×	×
[42]	Smart building	Queueing network	MRT, resource utilization, flow rate, discard rate, and the number of messages	×	×	×
[49]	Smart building	Petri net	Availability and performance	×	×	×
[36]	Smart hospital	Petri net	Availability and performance	×	×	×
[40]	Smart hospital	Petri net	MRT, resource utilization, and discard rate	×	×	×
[46]	Smart hospital	Queueing Network	MRT, utilization, throughput, number of messagese, and drop rate	×	×	×
[19]	Smart surveillance	Petri net	Availability and reliability	×	×	×
[39]	VANETs	Petri net	Number of traffic accident prediction	×	×	×
[15]	VANETs	Petri net	Computational time	×	×	×
[41]	VANETs	Petri net	MRT, resource utilization, and the number of requests discards	×	×	×
[5]	VANETs	Petri net and RBD	Availability and reliability	×	×	×
This work	Smart building, smart surveillance and VANETs	Fault tree and Markov chain	Availability and reliability	✓	✓	✓

While the existing literature has predominantly adopted the Petri net approach, our proposal offers a more comprehensive and holistic perspective on the dependability of smart city architectures. In future research, we plan to investigate the impact of different distribution types on the results, establish a more realistic baseline scenario, conduct performance analysis to evaluate the impact of availability on response time and throughput, and incorporate additional components such as onboard units of vehicles in vehicular ad hoc networks (VANETs) to further enhance the realism of our model. The field of Internet of Things (IoT) and dependability has been enriched by several notable studies, including Silva et al. [47], Ojie et al. [37], Andrade et al. [3], and Boano et al. [8]. Silva et al. [47] propose a tool for evaluating the dependability of IoT applications, considering both hardware faults and permanent link faults.

Similarly, Ojie et al. [37] explore dependability issues in IoT applications through a comprehensive analysis of existing literature and conceptualization of dependability requirements, regardless of the size and domain of the applications. Andrade et al. [3] present a Petri net-based approach for modeling and analyzing disaster recovery (DR) solutions for IoT infrastructures. Their proposed models enable assessment of important DR measures such as system availability, cost, and recovery time objective, with a real-world healthcare IoT system used as a case study. Boano et al. [8] provide methods and tools for predicting, guaranteeing, and enhancing the dependability of IoT systems. They highlight contributions in dependable wireless networking and present a cost-effective solution for ensuring IoT applications meet specific performance requirements despite challenges posed by low-power wireless networks and their environments. It is noteworthy that these studies focus on specific aspects of networking, which may differ from the broad perspective of our proposal in modeling smart city systems.

3 Background

This section presents some concepts regarding modeling mechanisms.

3.1 Modelling with Fault Tree

In a Hierarchical Fault Tree (HFT) analysis, the system under study is modeled as a tree-like structure, with the top-level representing the overall system failure and lower levels representing the failures of individual components that contribute to the overall system failure. Fault Tree Analysis (FTA) is a method for modeling the top-down failure behavior of a system by identifying and analyzing the combinations of basic events that can lead to a specific top event, such as system failure. The basic events are typically low-level components or subsystems that can fail, while the top event represents the overall system failure. In an HFT, the top-level represents the system-level failure and each lower level represents a more detailed failure mode. Each level of the fault tree is constructed using basic events, which are the lowest level failure modes, and gates, which are logical operators that combine the basic events. The mathematical representation of a fault tree is a Boolean function, represented by a set of Boolean equations. The probability of the top event can be calculated by using the probability of the basic events and the logical operators connecting them. Fault Tree Analysis (FTA) is a technique used to identify and assess potential failure modes and their causes in a system. It is often used in conjunction with a Reliability Block Diagram (RBD) to identify the cause-and-effect relationships between different components and subsystems in a system. A Fault Tree is a graphical representation of the logical relationships between the various failure events that can lead to a specific top event, such as system failure. The basic building blocks of a Fault Tree are the basic events, which represent a single failure or malfunction of a component, and the gates, which represent the logical relationships between the events. The probability of the top-level system failure, $Pr(S)$, can be calculated using the following equation:

$$Pr(S) = 1 - Pr(S') = 1 - \prod (1 - Pr(E_i)) \tag{1}$$

Where $Pr(S')$ is the probability of the system not failing, $Pr(E_i)$ is the probability of a basic event i, and the product is taken over all basic events.

In summary, Fault Tree Analysis and Markov Chain are two methodologies for modeling the dependability of systems. While FTA is used for modeling top-down failure behavior, MC is used for modeling the time-dependent behavior of a system. Both methodologies can be used in hierarchical manner to represent the system behavior in more detail. Combining the use of fault tree and Markov chain analysis in a hierarchical manner allows for a more detailed and accurate assessment of system dependability and performance.

Let X be a random variable which represents the lifetime of a device. That is, if the device is turned on at time zero, X would represent the time at which the device fails. The reliability function of the device, RX (t), is simply the probability that the device is still functioning at time t:

$$Rx(t) = Pr(X > t) \tag{2}$$

Where $R(t)$ is the reliability of the system at time t, and $Pr(failure)$ is the probability of failure at time t. Additionally, we would like to note that the reliability of a system can also be represented by the mean time between failures (MTBF) or the mean time to failure (MTTF). MTBF is calculated as the total operating time of a system divided by the number of failures, while MTTF is calculated as the reciprocal of the failure rate. These formulas are as follows:

$$\begin{aligned} MTBF &= \frac{Total\ Operating\ Time}{Number\ of\ Failures} \\ MTTF &= \frac{1}{Failure\ Rate} \end{aligned} \tag{3}$$

Availability (A) is a measure of the system's ability to perform its intended function when required. It is typically expressed as a percentage and is calculated using the formula:

$$A = \frac{uptime}{uptime + downtime} \times 100\% \tag{4}$$

Where uptime is the total time the system is functioning as intended and downtime is the total time the system is not functioning as intended. This formula can also be expressed in terms of Mean Time Between Failure (MTBF) and Mean Time To Repair (MTTR) as:

$$A = \frac{MTBF}{(MTBF + MTTR)} \tag{5}$$

In this equation, MTBF represents the average time between failures and MTTR represents the average time required to repair the system after a failure. It's also important to note that the availability of a system can be affected by factors such as concurrent failures, maintenance, and scheduled downtimes, which should be considered when calculating availability.

3.2 Modelling with Markov Chain

Markov models are the fundamental building blocks upon which many quantitative analytical performance techniques are built [26, 33]. Such models may be used to represent the interactions between various system components, for both descriptive and predictive purposes [35]. Markov models have been in use intensively in performance and dependability modeling since around the fifties [13]. Besides computer science, the range of applications for Markov models is very extensive. Economics, meteorology, physics, chemistry and telecommunications are some examples of fields which found in this kind of stochastic[1] modeling a good approach to address various problems.

A Markov model can be described as a state-space diagram associated to a Markov process, which constitutes a subclass of stochastic processes. A definition of stochastic process is presented: A stochastic process is a family of random variables $\{X_t : t \in T\}$ where each random variable X_t is indexed by parameter $t \in T$, which is usually called the time parameter if $T \subset R_+ = [0, \infty)$, i.e., T is in the set of non-negative real numbers. The set of all possible valuesof X_t (for each $t \in T$) is known as the state space S of the stochastic process [9].

Let $Pr\{k\}$ be the probability of a given event k occurs. A Markov process is a stochastic process in which $Pr\left\{X_{t_{n+1}} \leq s_{i+1}\right\}$ depends only on the last previous value X_{t_n}, for all $t_{n+1} > t_n > t_{n-1} > \cdots > t_0 = 0$, and all $s_i \in S$. This is the so-called Markov property [20], which, in plain words, means that the future evolution of the Markov process is totally described by the current state, and is independent of past states [20].

In this work, there is only interest on discrete (countable) state space Markov models, also known as Markov chains, which are distinguished in two classes: Discrete Time Markov Chains (DTMCs) and Continuous Time Markov

[1] Stochastic refers to something which involves or contains a random variable or variables

Chains (CTMCs) [25]. In DTMCs, the transitions between states can only take place at known intervals, that is, step-by-step. Systems where transitions only occur in a daily basis, or following a strict discrete clock are well represented by DTMCs. If state transitions may occur at arbitrary (continuous) instants of time, the Markov chain is a CTMC. The Markov property implies that the time of transitions is driven by a memoryless distribution [9]. In the case of DTMC, the geometric distribution is the only discrete time distribution that presents the memoryless property. In the case of CTMC, the exponential distribution is used.

Markov chains can be represented as a directed graph with labeled transitions, indicating the probability or rate at which such transitions occur. When dealing with CTMCs, such as the availability model of Figure 1, transitions occur with a rate, instead of a probability, due to the continuous nature of this kind of model. The CTMC is represented through its transition matrix, often referenced as infinitesimal generator matrix. Considering the CTMC availability model of Figure 1, the rates are measured in failures per second, repairs per second, and detections per second. The generator matrix Q is composed by components q_{ii} and q_{ij}, where $i \neq j$ and $\sum q_{ij} = -q_{ii}$. Using the availability model that was just mentioned, considering a state-space S = {Up,Down,Repair} = {0,1,2} the Q matrix is:

$$Q = \begin{pmatrix} q_{00} & q_{01} & q_{02} \\ q_{10} & q_{11} & q_{12} \\ q_{20} & q_{21} & q_{22} \end{pmatrix} = \begin{pmatrix} -0.001 & 0.001 & 0 \\ 0 & -2 & 2 \\ 0.2 & 0 & -0.2 \end{pmatrix}$$

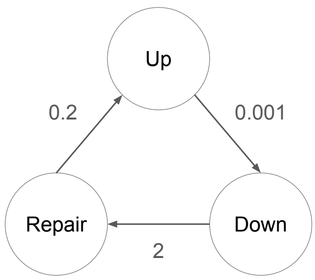


Fig. 1: Simple CTMC

Equation 6 and the system of Equations 7 describe the computation of the state probability vector, respectively for transient (i.e., time-dependent) analysis, and steady-state (i.e., stationary) analysis. From the state probability vector, nearly all other metrics can be derived, depending on the system that is represented.

$$\underline{\pi}'(t) = \underline{\pi}(t)Q, \qquad given\ \underline{\pi}(0) \tag{6}$$

$$\underline{\pi}Q = 0, \sum_{i \in S} \pi_i = 1 \tag{7}$$

Detailed explanations about how to obtain these equations may be found in [21, 9]. For all kinds of analysis using Markov chains, an important aspect must be kept in mind: the exponential distribution of transition rates. The behavior of events in many computer systems may be fit better by other probability distributions, but in some of these situations the exponential distribution is considered an acceptable approximation, enabling the use of Markov models. It is also possible to adapt transition in Markov chains to represent other distributions by means of phase approximation, as shown in [33]. The use of such technique allows the modeling of events described by distributions such as Weibull, Erlang, Cox, hypoexponential, and hyperexponential.

4 Methodology

This section presents the methodology employed in this study, outlining the steps taken to improve the availability and reliability of the system under examination. The primary objective of this section is to facilitate future replication of the methodology, enabling the enhancement of dependability for systems with similar physical and logical structures. The methodology describes the processes used in constructing the proposed environment, including the evaluation and construction of models of the infrastructure used by the system, as well as the proposal of alternative structures aimed at increasing the system's dependability. Figure 2 provides a visual representation of the steps in the proposed methodology, with the processes depicted as macro-activities, which are described in detail below.

System Understanding. This process involves understanding the structure of the system, including the study of the main physical and logical components, their basic functionalities, and the existing components in the physical and virtual layers, among other characteristics of the environment. The evaluated system was designed based on information from the literature. The decision on which macro-components to include was made first. The three macro-components (VANET, Smart Building and Surveillance) were chosen as they are common in many works related to smart cities [34, 43, 40, 19, 5]. Additionally, the availability of parameters to feed the entire proposed model was considered to be an important factor. **Architecture Design**. Before actually modeling the system, it is necessary to define the specific components and how they relate to one another. This step involves determining the number of sub-components each macro-component has and defining the failure requirements of the system. **System Modeling**. The main objective of this process is to construct a model that accurately represents the studied environment, classified as a baseline, allowing for more precise analysis of the environment and proposals for improvements in dependability. The proposed models were designed and evaluated using the Mercury [44] tool. Two types of models are

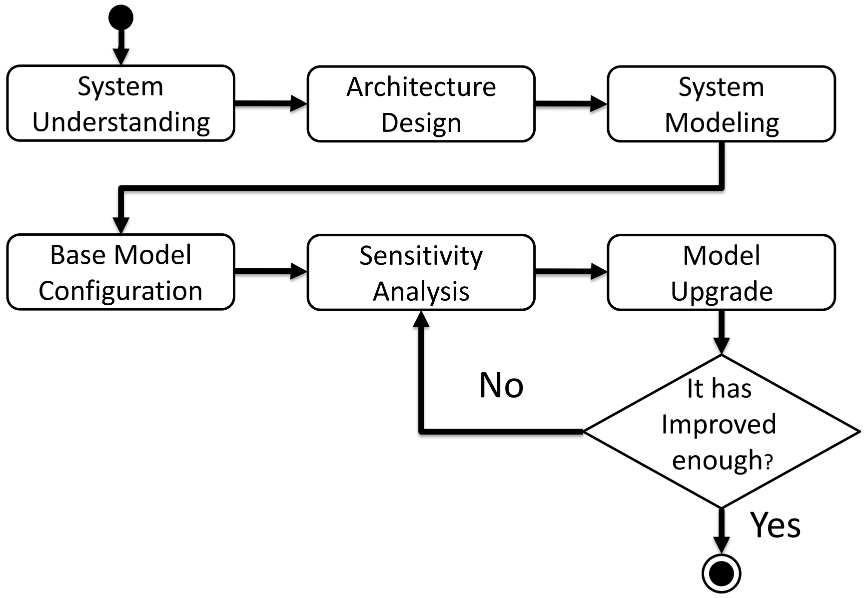


Fig. 2: Modelling and Analysis Methodology Organized in Macro-Activities.

proposed in this work: fault tree and Markov chains. In this step, for example, the gates of the fault tree were specified. **Base Model Configuration**. In this step, the model receives the values of the availability parameters of each of the chosen components, as well as information on which components will have redundancy and the number of redundant elements (referred to as KooN).

Sensitivity Analysis. The sensitivity analysis process aims to identify the components that can be adjusted in order to increase the dependability of the environment. This activity aims to adjust the environment with the fewest possible changes, allowing for more assertive changes. In this work, the new concept of sequential sensitivity analysis (SSA) is employed. SSA involves not only one, but multiple executions of sensitivity analysis, allowing for even more effective upgrades in the designed models. **Model Upgrade**. The main objective of this macro-activity is to construct a model with better dependability results for the studied environment. This construction is linked to the sensitivity list, taking as input one or more components that will be improved. Several techniques are used to improve availability and reliability. In this work, changes in gates (in fault trees) and changes in the structure of Markov chains are proposed.

In this work, a specific scenario of simple infrastructure and manual configurations was considered for modeling and analyzing the dependability of such systems. The proposed methodologies provide a clear and easy-to-understand approach for this specific scenario. However, it is important to note that there are more advanced techniques, such as the automatic build of multiple/hierarchical formalism, that have been proposed in the literature for modeling and analyzing more complex systems. It should be noted that the model does not include dynamic aspects of the network, and the mean time to failures (MTTF) and mean time to recovery (MTTR) of all components are calculated from more detailed sub-architectures.

For example, the cloud component can have near infinite possibilities of granularity. While it was not the goal of this work to detail such components, the proposed model can still provide a reasonably big picture of the modelled system. It is worth mentioning that the literature does not present a modelling approach combining distinct contexts as presented in this work. Focusing on a specific context allows for a more accurate representation of network, power source, mobility, etc. In this sense, the proposed model can be considered as a complementary approach that can be used in specific scenarios, rather than a replacement for more advanced techniques.

5 Architecture

Figure 3 presents an overview of a computer systems cluster in a smart city. The architecture is designed to separate and emphasize three crucial sub-architectures: Smart Surveillance, VANETs, and Smart Building. Each of the sub-architectures has distinct hardware and software requirements. At the core of the architecture, elements that are shared by the three sub-architectures are highlighted. The following sections will provide a detailed description of the four macro architectural components.

5.1 Smart Surveillance

The use of Closed-Circuit Television (CCTV) has seen a significant increase in the past two decades, particularly following the 9/11 attacks. The United Kingdom, considered the most monitored country in the world, has approximately 4 million surveillance cameras, with an estimated 300 cameras capturing footage of a London resident daily [27]. These cameras cover both public and private areas, including downtown areas, airports, and shopping malls. This growth is driven by two factors: the decrease in the cost of CCTV systems and the increasing demand for security. The purpose of an intelligent video surveillance system is to detect anomalies, or behavior that does not conform to a normal scene behavior specification. Smart surveillance plays an important role in anomaly detection by extracting, representing, and interpreting visual information for learning and classifying behavior. There have been numerous works published in the area of intelligent surveillance, with some studies aiming to determine the level of system supervision, extracting characteristics that describe the behavior of different classes of objects in the scene, and generating

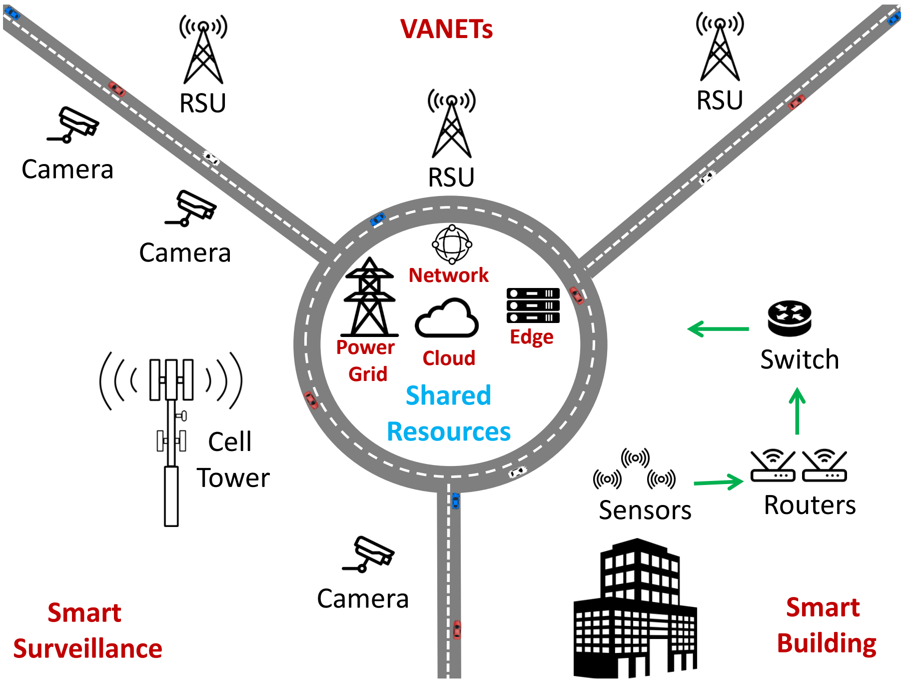


Fig. 3: Smart City Systems Architecture.

a compact representation of the extracted characteristics. All these practical applications require high computational capacity for processing, storage, and communication.

To effectively cover a large city with a multi-camera scenario (fixed and mobile), a mobile network capable of handling the large amount of data is required. In addition to providing characteristics such as broadband, reliability, latency, and mobility superior to previous networks, it must also be highly adaptable. Such a network must be fast, responsive, and stable, which are characteristics offered by 5G. 5G can adapt to fluctuations in traffic, dynamically directing services, such as a higher data transfer rate, if a specific application or user requires it. Therefore, the smart surveillance sub-architecture includes a set of high-performance cameras and one or more transmission towers. The transmission towers (cell towers) then communicate with centralized edge and cloud resources.

5.2 VANETs

The increase in population concentration in urban centers, particularly in large metropolitan regions, and the resulting mobility linked to the increased use of motor vehicles have negative impacts on society. These factors contribute to poor quality of life for drivers and pedestrians, as well as environmental damage associated with carbon dioxide emissions and economic losses. Congestion is one of the most severe issues in large cities worldwide. There is significant research being conducted to address this problem, and one promising approach is smart mobility, which is a domain of smart cities. The main idea behind smart mobility is the use of Information and Communication Technologies (ICT) to enhance the use of transit infrastructure in cities. Many smart mobility initiatives are already in place, such as VANETs networks.

Vehicle networks allow for the exchange of information between vehicles (V2V - Vehicle-to-Vehicle communication) and between vehicles and infrastructure located on the side of the roads (V2I - Vehicle-to-Infrastructure communication), with the possibility of utilizing both types of communications on the same network (V2X architecture). Vehicular networks have unique characteristics that present challenges in communication between vehicles, such as high node mobility and the potential for high rates of packet loss due to network congestion. Road Side Units (RSUs) are communication base stations in the vicinity of roads and highways that communicate with vehicles. These RSUs must communicate directly with remote resources such as edge servers and cloud computing services. Therefore, in the present architecture, we consider a set of RSUs that must be fully operational for the sub-architecture to be considered in an active state.

5.3 Smart Building

The Internet of Things (IoT) has great potential for use in a variety of applications, including commerce, industry, and education. IoT devices can be used, for example, to automate activities in smart homes. Identifying abnormal behavior in a monitored environment using IoT has proven to be very useful in residential settings. Historically, smart homes were primarily used to control environmental systems such as lighting and heating. However, the use of IoT has allowed for all electrical components in a home to be connected to a single system. A smart home can provide context-based services to homeowners using sensors and other technologies. From the concept of smart homes, IoT has been extended to the context of smart buildings. Due to their size and complexity, buildings require more sophisticated computing infrastructures and preparation to meet the high demand for data generation. Energy consumption is a major concern in smart building resource management. Intelligent energy management of a building can play a crucial role in addressing energy consumption, which has a significant impact on the environment. Most current IoT sensors must be installed throughout the building to collect data, which is then processed locally and remotely (edge/cloud). In the present work, we consider that each building environment (floor) has one or more routers, and the entire building has a switch device to forward data to the external environment.

5.4 Shared Resources

The proposed architecture for a smart city includes four shared components: power grid, cloud, edge, and network. These components are essential for the functioning of the smart city's three sub-architectures: Smart Surveillance, VANETs, and Smart Building. The power grid supplies electrical power for all devices, while the network connects the sub-architectures to the edge and cloud components. The edge and cloud play a crucial role in the system, with the edge providing faster information processing and the cloud handling more complex processing and storage for backup. It is important to note that these four shared components, along with the three sub-architectures, are of equal importance for the dependability of the overall smart city system. However, care must be taken to ensure proper consideration of their individual roles and effects on dependability.

6 Model

Figure 4 presents the fault tree model proposed in this work.

We observe two metrics in this paper: availability and reliability. Availability is the ability of a component to be in a functional state to perform a required function. Reliability is the ability of the equipment to satisfactorily perform a given function during a specific time.[32] All the models and respective solutions were performed using the modelling tool Mercury [44]. We tried to design a model well aligned with the architecture described in the previous section. The primary gate is an OR gate. For the model root (smart city system) to fail, all shared, or non-shared components must fail. For unshared resources to fail, one of the following components must fail: surveillance, smart building, or VANET.

In the proposed architecture, the cameras (CAs) or the tower (TO) are critical components for the surveillance sub-architecture. The failure of these components can cause the entire sub-architecture to fail. To account for this, a new type of gate, known as K out of N (KooN) gate, is introduced. The KooN gate is a practical tool for modeling component dependencies, where K out of N items must fail for the component to fail. For example, in the Figure 5, if there are five cameras in total and the model states `"3 out of 5 cameras"`, the component will only fail if three or more cameras fail. The proposed model includes five KooN gates that were strategically placed to accurately reflect the reality of modern cities. However, some simplifications were made to facilitate data manipulation. For example, the model assumes that there is only one tower in the surveillance module, as it is typically sufficient for coverage in city centers. The use of KooN gates also considered the likelihood of components having a shorter time to failure.

The smart building component comprises three sub-components: sensors (SS), routers (RTs), and a switch (ST). While other physical devices could have been broken down, such as actuators, we aim to simplify the architecture by focusing on more commonly adopted components. A K-out-of-N (KooN) gate

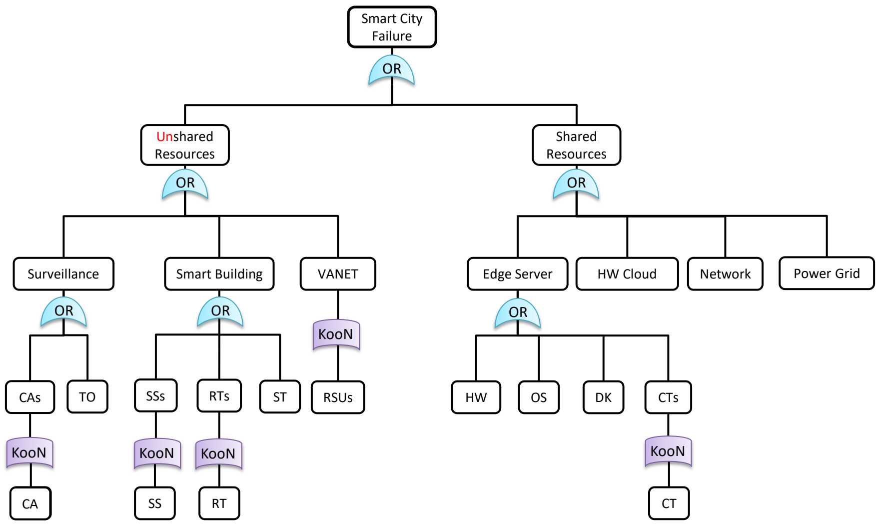


Fig. 4: Smart City Systems Fault Tree Model.

is applied for sensors and routers to account for the grouping of devices within a building. For example, in a smart hospital, sensors are typically organized by floor, with a specific number of sensors communicating with a router on each floor. The use of KooN gates also allows for more flexible definition of reliability requirements. Given the large number of sensors typically present in a building, it is unlikely that the failure of just one sensor would cause the entire system to fail.

The number of routers will be proportional to the number of monitored floors in the building, with all routers connected to a strategically placed switch. In the proposed architecture, the VANETs component is comprised of a single sub-component: Road Side Units (RSUs). Typically, VANETs consist of two basic elements: RSUs and Onboard Units (OBUs). However, due to the complexities associated with defining the actual quantity of OBUs in a smart city, they were not included in this study. The architectural planning of RSUs is relatively straightforward as they are fixed, pre-established structures. As with the sensors in the smart building component, it is not reasonable to assume that the failure of a single RSU would cause the entire system to fail, as RSUs are typically organized in a redundant manner.

The gate related to shared resources includes four macro-components: edge server, hardware cloud, network, and power grid. The edge server is divided into four sub-components: hardware (HW), operating system (OS), docker (DK), and containers (CTs). The use of Docker was incorporated to more accurately reflect modern and scalable architectures that utilize containers. The container component includes a KooN gate with a set of N parallel

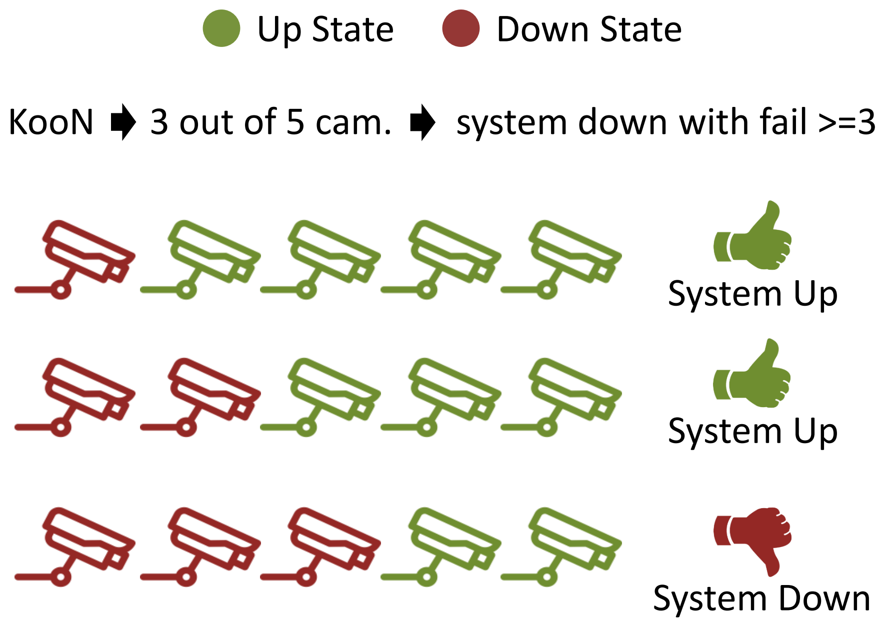


Fig. 5: KooN example using cameras in up and down states.

containers. The cloud architecture, in terms of software, can vary depending on the provider adopted. Therefore, the cloud component has been simplified to consider only the hardware aspect. The network component encompasses all wired communication interconnecting the smart city network. Lastly, the power grid component supplies power to the entire architecture under consideration. Renewable energy sources can also be considered in a subsequent phase, however, they are not included in this analysis due to the low failure rate of power grids. Table 2 presents the mean time to fail (MTTF) and mean time to repair (MTTR) for each component as extracted from literature sources [34, 43, 40, 19, 5].

6.1 Sensitivity Analysis

In this part of the work, we present a sensitivity analysis of the macro components of the model: surveillance (SV), smart building (BU), VANET (VN), edge server (ED), hw cloud (CL), network (NE) and power grid (PG). The objective is to identify which component has the greatest impact on the availability metric. Sensitivity analysis measures the local effect of a given input data on the output data. The main objective is to identify the weak links of computer systems and, subsequently, to adopt a set of techniques to improve such systems in different scenarios [11]. Sensitivity analysis can provide the necessary security and present the results within the perspective established by

Table 2: Model Parameters

Component		MTTF	MTTR
Surveillance	**Camera (CA)**	90000.00	1.00
	Cell Tower (TO)	150000.00	5.00
Smart Building	**Sensor (SS)**	300000.00	1.00
	Router (RO)	480770.00	8.00
	Switch (ST)	698220.00	8.00
Edge Server	**Hardware (HW)**	8760.00	1.67
	Operating System (OS)	1667.00	0.25
	Docker (DK)	2516.00	0.25
	Container (CT)	1258.00	0.23
Power Grid		8757.00	4.80
Hardware Cloud		8760.00	1.60
RSU		500.00	2.00
Network		83220.00	12.00

the system administrators. There are various methods available for performing sensitivity analysis, such as regression analysis, perturbation analysis (PA), one-to-one variation, Monte Carlo simulation, parametric differential analysis, correlation analysis, and percentage difference. The decision of which method to use is a challenging step, requiring the consideration of available computational resources and the characteristics of the addressed problems [12] [38].

The percentage difference method was chosen to perform the sensitivity analysis of the fault tree model, as it is an approach that does not require the use of a continuous domain for the input values of parameters. This process is based on calculating the percentage difference when varying an input parameter from its minimum value to its maximum value. It is necessary to use the entire range of possible values of each parameter in order to calculate the sensitivities of the parameters, as outlined in Equation 8. The expression *max Y(θ)* and *min Y(θ)* represent the maximum and minimum output values, respectively, as measured by varying the parameter θ over a range of its n possible values of interest. If it is known that *Y (θ)* varies monotonically, only the extreme values of θ (i.e., θ_1 and θ_n) need to be used to calculate *max Y(θ)* , *min Y(θ)* and as a result S_θY [16, 22].

$$S_\theta\{Y\} = \frac{max\{Y(\theta)\} - min\{Y(\theta)\}}{max\{Y(\theta)\}} \tag{8}$$

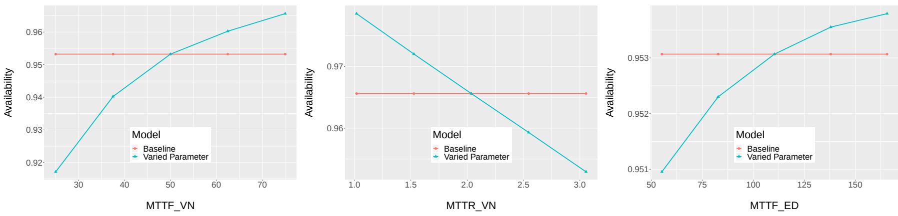
We present the calculation of the sensitivity index by varying one parameter over its design range. It is important to note that the values of the remaining parameters are held constant during this calculation. This approach allows us to isolate the effect of a single parameter on the overall system performance and assess its relative importance. Additionally, it should be noted that in the sensitivity analysis, a set of scenarios were considered with different values of the remaining parameters to ensure the robustness of the results. The sensitivity index provides a measure of the relative importance of each design parameter and allows the designer to prioritize design decisions.

Table 3 presents the sensitivity indexes in descending order. Therefore, the three most impacting components are: VANETs, edge server and smart building. VANETs is the most influential component probably due to its low MTTF compared to the other components.

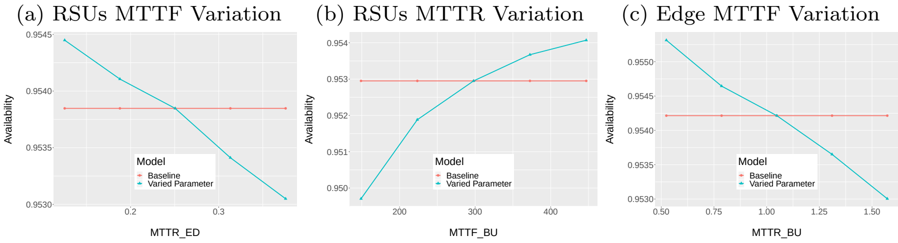
Table 3: Sensitivity Analysis Result.

Parameter	Sensitivity Index
MTTF_VN	0.0258777602608578
MTTR_VN	0.01331305761385408
MTTF_BU	0.002206516444554778
MTTF_ED	0.0016366888322117829
MTTR_BU	0.0010814511382172705
MTTR_ED	7.069677847551544E-4
MTTF_SV	6.40961033860294E-4
MTTF_PG	3.216745273348131E-4
MTTR_PG	3.170190776072599E-4
MTTR_SV	2.72584548561388E-4
MTTF_CL	2.2045099731459627E-4
MTTR_NE	1.0141813678947335E-4
MTTF_NE	8.385117185474646E-5
MTTR_CL	2.158935723395704E-5

Figure 6 presents the parameters variation for each eminent component. The variation have considered increments and decrements of 25% from the baseline. It is important to note that edge and smart building varies the availability inside the range of 95%. However, the RSU MTTF goes from 92% to more than 96%.



(a) RSUs MTTF Variation (b) RSUs MTTR Variation (c) Edge MTTF Variation



(d) Edge MTTR Variation (e) Building MTTF Variation (f) Building MTTR Variation

Fig. 6: Parameters Variation of Macro Components

7 Case Studies

This section presents two case studies that aim to identify system availability and reliability for the different parameters. The case studies focus on the five KooN gates. Changing the KooN gate can represent two aspects: a change in the failure requirement and a change in the number of available resources. These two aspects were explored and described in the following sections.

7.1 Scenario One - Changing the Failure Requirement

The first aspect of the KooN gate change pertains to the variation of the k factor, without altering the value of N. This variation indicates a change in the failure requirements of that component. As the value of k increases, the component becomes less likely to fail, as there will be more redundant elements. In this scenario, three configurations of the KooN gates were applied, as detailed in Table 4. The settings for the five components with a KooN gate were changed: cameras, sensors, routers, RSUs, and containers. The total capacity for each component was fixed, with 100 cameras, 1000 sensors, and 10 routers, RSUs, and containers. For the configurations C1, C2, and C3, three different k values were adopted, respectively: k=1, k=N/2, and k=N.

Table 4: Configurations considering distinct K values.

Components	Configurations		
	C01 - 1 oo N	**C02 - N/2 oo N**	**C03 - N oo N**
Cameras	1 out of 100	50 out of 100	100 out of 100
Sensors	1 out of 1000	500 out of 1000	1000 out of 1000
Routers	1 out of 10	5 out of 10	10 out of 10
RSU	1 out of 10	5 out of 10	10 out of 10
Containers	1 out of 10	5 out of 10	10 out of 10

The first aspect of the KooN gate change is related to the variation in the k factor (without changing N). This variation indicates a change in the failure requirement of that component. As the k value increases, the system becomes less likely to fail, as there will be more redundant elements. In this scenario, three configurations of the KooN gates were applied, as detailed in Table 4. The total capacity of the cameras, sensors, routers, RSUs, and containers were fixed at 100, 1000, 10, respectively. The steady-state availability results for these configurations are presented in Figure 7. Configuration C1 had an availability of approximately 95.3%, while C2 and C3 had an availability of approximately 99.8%. Configuration C1 can be considered as a very demanding failure requirement, as the failure of just one out of a thousand sensors would cause the system to fail. However, it serves to illustrate the ease of exploration of the model. If a Service Level Agreement (SLA) with a target availability of 98% is established, configurations C2 and C3 would meet this demand equally. However, C3 is more fault-tolerant as it requires all elements to fail for the system to enter a failed state. Therefore, it is important to consider the reliability metric when defining an SLA.

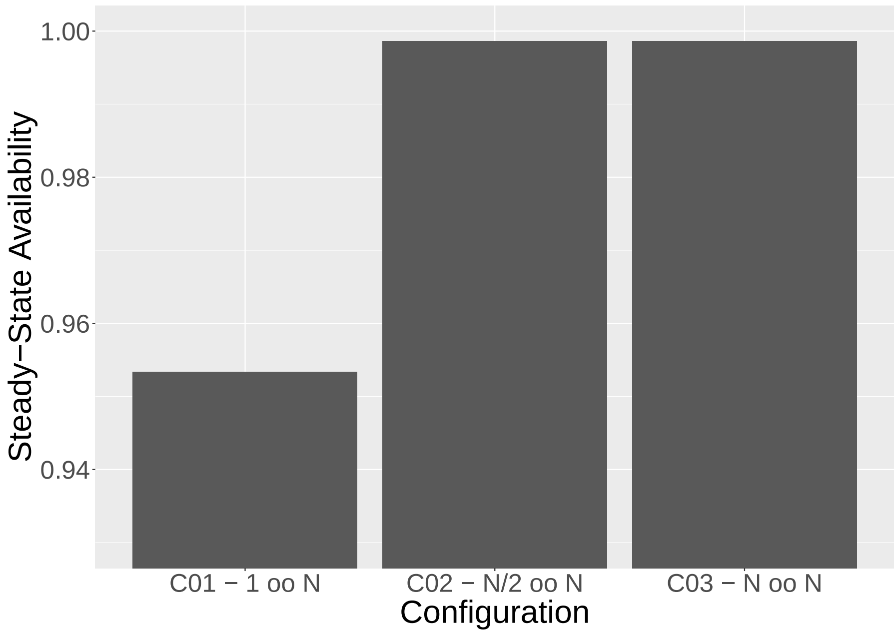


Fig. 7: Scenario One - Variation of KooN observing availability.

The Figure 8 illustrates the results of the reliability analysis for the three configurations of the KooN gates, C1, C2, and C3. The x-axis represents the elapsed time since the system is considered active, with the simulation considering a time of 2000 hours. It is evident that configuration C3 has better reliability than C2, which in turn has better reliability than C1, for the majority of the time. At time zero, the system's reliability is 100%, as the system has not yet attempted to fail. As time progresses, reliability is diminished. Initially, C2 and C3 are practically the same until 125 hours, after which they begin to diverge. C1, with its high constraint, achieves zero reliability near 250 hours (a little over a week). C2 reaches zero reliability after 500 hours, while C3 reaches zero reliability only after 2000 hours (over two months).

The Figure 7 and 8 present the results of a sensitivity analysis performed on the fault tree model using the percentage difference method. The analysis aimed to identify the impact of varying the k factor of the KooN gates on the availability and reliability of the system. The results demonstrate that increasing the value of k, or the number of redundant elements, results in a higher availability and reliability for the system. The study also highlighted the importance of considering the time frame when defining Service Level

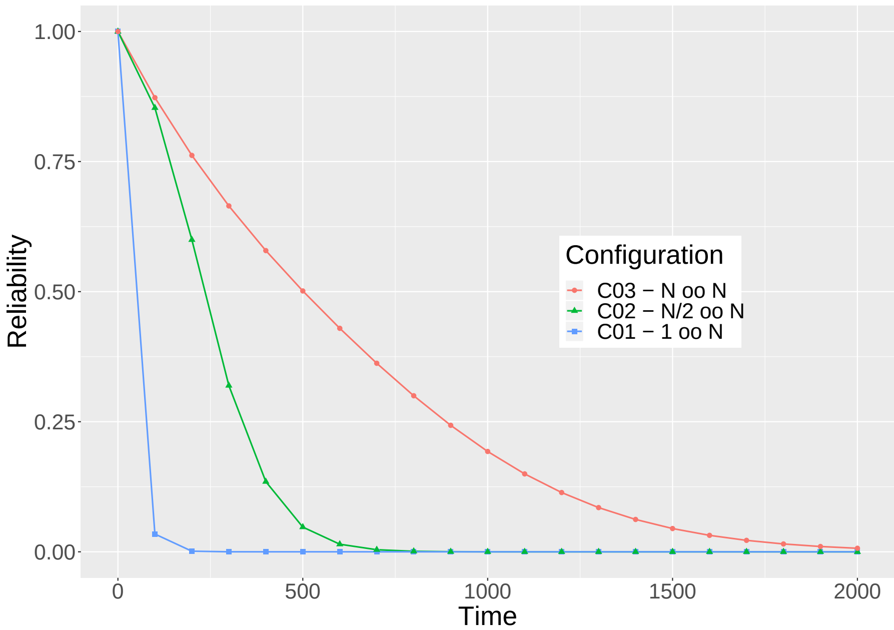


Fig. 8: Scenario One - Variation of KooN observing reliability.

Agreements (SLAs) for reliability. As the results can vary greatly depending on the time analyzed, it is important to carefully consider the consequences of partial unavailability when defining SLAs with high k values. Additionally, it is also important to take into account the maintenance of the system, as high k values may not trigger corrective actions for individual failures and may negatively affect the system's performance.

7.2 Scenario Two - Changing the Quantity of Available Resources

The Vehicle Ad-hoc Network (VANET) was identified as the most significant macro-component in the sensitivity analysis. In this case study, we focus on the VANET component by altering the respective KooN gate. This scenario involves variations in both the value of K and the value of N. It is important to note that when K is equal to N, there is only failure if all elements fail, referred to as the "maximized failure condition." Additionally, it is worth noting that N represents the number of available elements. In this scenario, we aim to maintain the maximized failure condition by changing the number of available devices (N) as outlined in Table 5. We simulated the increase in available Road Side Units (RSUs) with three possible configurations.

As shown in Figure 9, the system's availability as a whole for the variation described above is depicted. In comparison to the previous scenario, the impact on availability is relatively small as the configuration changes. With two

Table 5: Configurations considering distinct K and N values for VANET component.

	Configuration		
	C1	**C2**	**C3**
KooN of RSU	2 out of 2	5 out of 5	10 out of 10

available RSUs, the availability is recorded as `A = 99.8625885`%. The C2 and C3 settings have identical values: `A=99.8641736`%. There is, therefore, very little difference in changing the settings for availability. However, the difference represents an hour of downtime. Downtime is calculated using the equation D = (1 - A) $\times$ 8760, where A denotes availability and 8760 represents the number of hours in a year. Although small, this one hour of downtime can potentially result in traffic accidents.

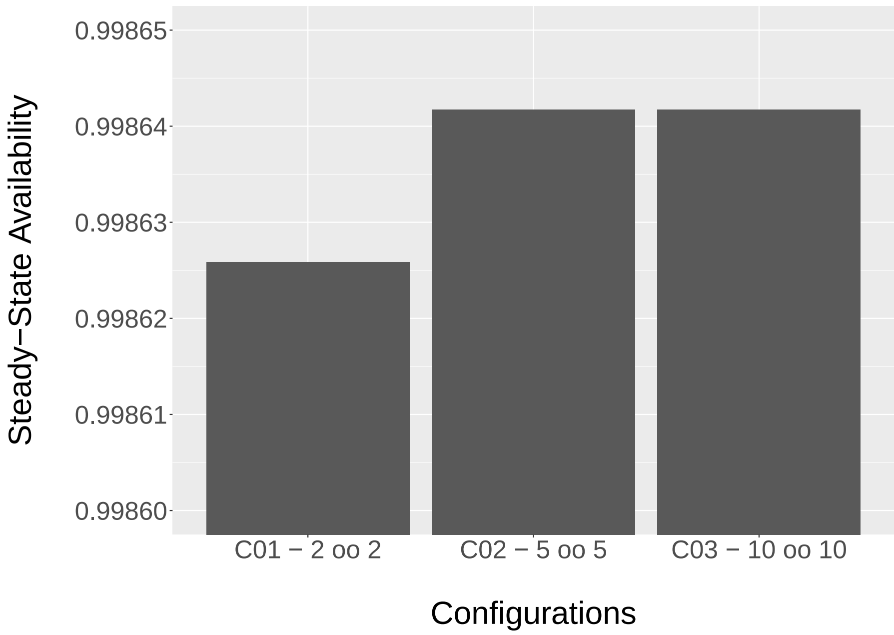


Fig. 9: Availability changing K and N of RSU.

Figure 10 presents the results of the reliability analysis for the second case study, considering a maximum time of 2000 hours. Unlike availability, the three configurations exhibit differences in terms of reliability. Reliability is directly proportional to the number of RSUs. Therefore, overall, the C3 configuration has higher reliability than the C2 configuration, which in turn has higher reliability than the C1 configuration. Starting from time zero, the C2 and C3 configurations are quite similar. However, from point 500h onwards, the lines begin to diverge. At the point of greatest distance (approximately 750h), the C1 configuration has a reliability of 12%, the C2 configuration has a reliability of 25%, and the C3 configuration has a reliability of 37%. The C1 and C3 configurations exhibit a significant difference for half of the analyzed time. The lines converge only at point 2000h. Therefore, arbitrarily adding RSUs can be an effective strategy for improving reliability, but it may not have a significant impact on availability.

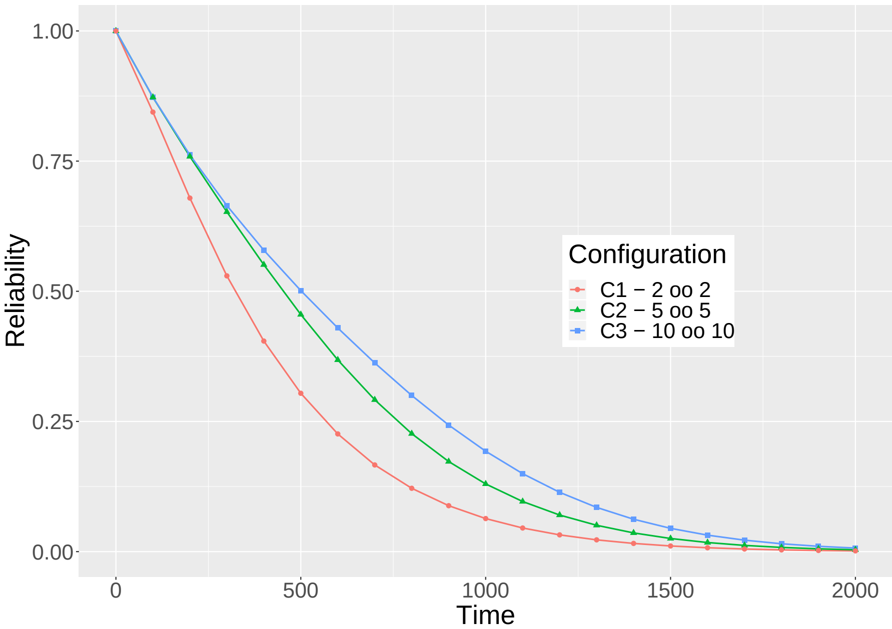


Fig. 10: Availability changing K and N of RSU.

7.3 Scenario Three - Changes Guided by Sensitivity Analysis

The design of smart city systems necessitates achieving minimum acceptable levels of Mean Time to Failure (MTTF). Attaining specific MTTF values in multi-component systems can be challenging. This scenario presents a methodology based on the successive use of sensitivity analysis to identify which components require modification to have the greatest impact on the system. The nature of the change to be applied depends on the characteristics of each component. Possible change options include the implementation of redundancy directly to the component, incorporating redundant elements into the infrastructure, or replacing an existing component with a more reliable one. Figure 11 illustrates the actions taken to achieve a system MTTF of 500 hours.

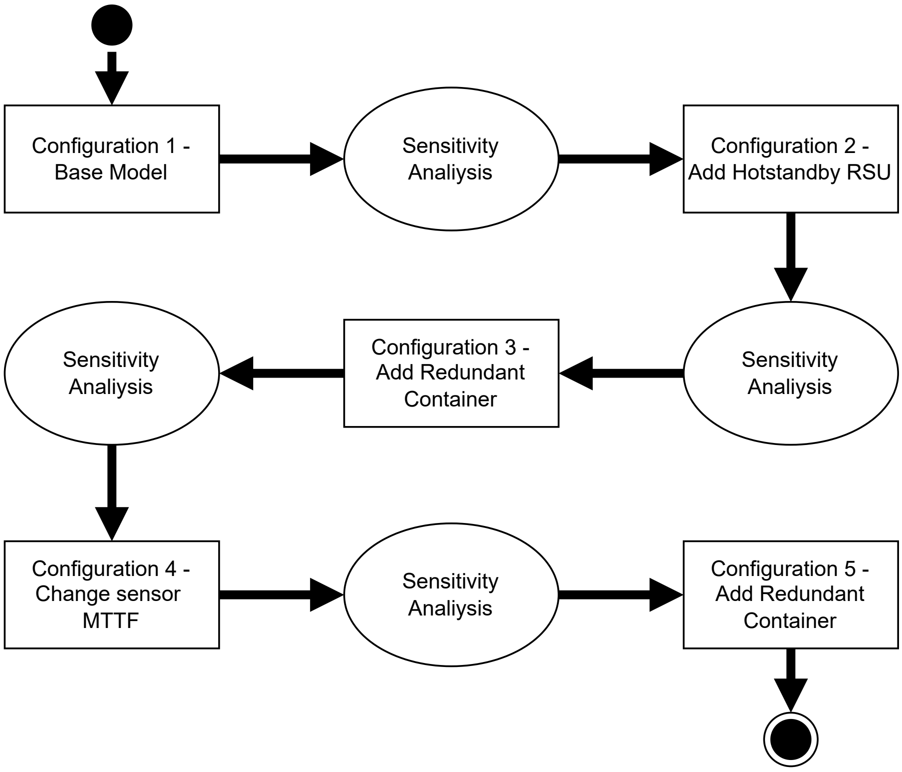


Fig. 11: Actions performed to achieve the minimum MTTF.

Initially, the MTTF of the base system was calculated as 80.72 hours. The base model was then subjected to sensitivity analysis based on the general system MTTF metric result. The outcome of this analysis can be observed in column Configuration 1 of Table 6, where it is evident that the Road Side Unit (RSU) is the most critical component. To improve this component, an action was taken to add an RSU in hot-standby mode, meaning that each RSU receives a redundant RSU as a secondary processing option.

A Continuous-Time Markov Chain (CTMC) model was developed to determine the MTTF of the RSU in hot-standby mode. Figure 12 illustrates the CTMC model for the RSU. This model comprises four states (UU, UD, DU, and DD) and utilizes a failure rate of λ for the failure of an RSU and a recovery rate of μ for the recovery of an RSU. The failure rate of an RSU, represented by λ, has a value of 1/500.00, and the repair rate of an RSU, represented by μ, has a value of 1/2.00, as per the parameters outlined in Table 2. The UU state represents the scenario where two RSUs from a single point are functioning correctly and can only fail at the λ rate. The UD and DU states represent the scenarios when one RSU has failed, but the other remains operational, in which

Table 6: MTTF sensitivity analysis with highest indexes colored in blue.

Parameter	Configuration 1	Configuration 2	Configuration 3	Configuration 4
rsu MTTF	0.12953794	8.52E-05	2.06E-04	3.18E-04
ct MTTF	0.040002993	0.118845515	0.060952061	0.102774879
sensor MTTF	0.009643628	0.036647992	0.071397819	0.002071902
os MTTF	0.007715903	0.016740269	0.026970735	0.033861797
dck MTTF	0.005132598	0.011189549	0.018099929	0.022789887
cloud MTTF	0.001510418	0.003443205	0.005489943	0.007096034
power grid MTTF	0.001510096	0.003439918	0.005482019	0.007082858
hardware MTTF	0.001472536	0.003205538	0.005185181	0.006527514
camera MTTF	0.001334406	0.006006158	0.012952436	0.019616187
network MTTF	1.59E-04	3.63E-04	5.79E-04	7.49E-04
cell tower MTTF	8.04E-05	1.84E-04	2.94E-04	3.82E-04
switch MTTF	1.74E-05	4.03E-05	6.52E-05	8.22E-05
router MTTF	4.96E-07	2.69E-06	6.59E-06	1.03E-05

case, as the system is still operational, it can recover at the μ rate. The DD state represents the scenario where both RSUs have failed, and the absorbing state is used to determine the MTTF of this Hot-standby RSU configuration.

The MTTF value for the RSU configuration in this scenario is 84166.67 hours, which is significantly higher than using a single RSU at each point, which is 500 hours. Incorporating this MTTF value into the system raised the overall MTTF to 178.93 hours, which is significantly higher than the previous value but still below the target stipulated in this use case. The Configuration 2 column of Table 6 presents the results of the new sensitivity analysis of the last configuration, and it was identified that the container is the most critical element. To address this, in addition to using the redundant RSU, it was decided to add new containers, as this is an action that can be carried out with relative ease and low cost. The Edge server received 14 new containers in a KooN configuration. This change increased the MTTF to 298.77 hours, but more changes were still needed to reach the desired goal. The sensitivity analysis of the system with additional redundant containers, as outlined in Configuration 3 of Table 6, identified that the sensors are the most critical elements.

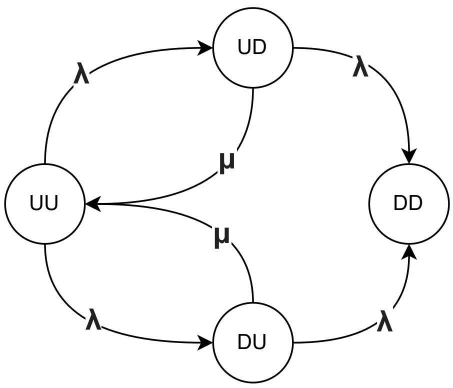


Fig. 12: Hot-standby at the RSU.

The availability of sensors was improved by increasing its MTTF by 10%. The outcome of this change resulted in an MTTF of 393.77 hours, which is still insufficient to meet the minimum required value. The subsequent sensitivity analysis of the entire system with this configuration can be found in Table 6 - Configuration 4. This result demonstrates that the increase in redundant containers was not sufficient to stop it from being a significant factor in the system's MTTF. Therefore, it was determined that the containers required a further increase in redundant components. Various quantities of containers were tested, and it was observed that the change from 14 to 21 redundant containers resulted in an MTTF of 505.29 hours, which exceeds the desired value for the system.

Figure 13 illustrates the MTTF for each different configuration. Five configurations were performed, resulting in an increase of the system's MTTF from 80.72 hours to 505.29 hours. As anticipated, the changes also impacted the system's reliability, which can be observed in the reliability graph shown in Figure 14. The impact on reliability can be observed by examining the impact of modifying the components with the highest sensitivity index in each configuration on reliability. For example, the system reliability at 400 hours in Configuration 1 is only 02%, while in Configuration 2, the reliability is 6.52%. On the other hand, Configuration 3 increases the reliability to 27.93%. In Configurations 4 and 5, the reliabilities are 45.09% and 53.37%, respectively. This case study illustrates the usefulness of the model combined with the use of sequential sensitivity analyses to increase the MTTF and reliability of systems with multiple components.

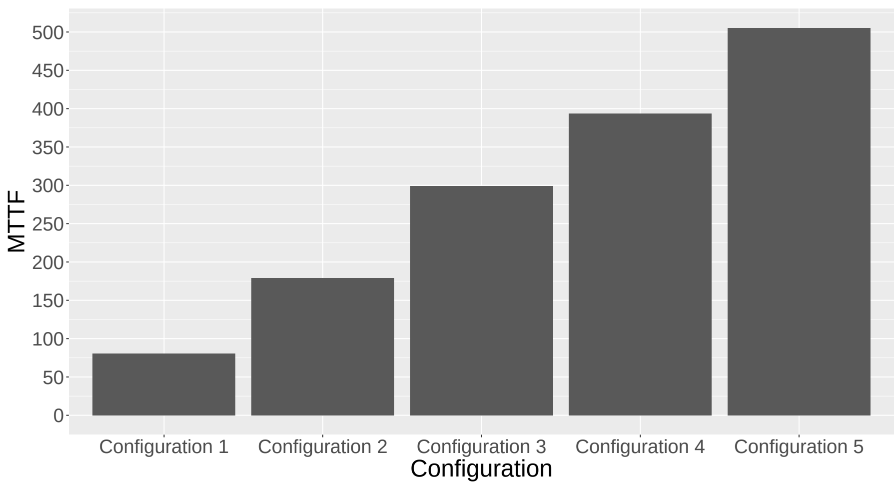


Fig. 13: System MTTF.

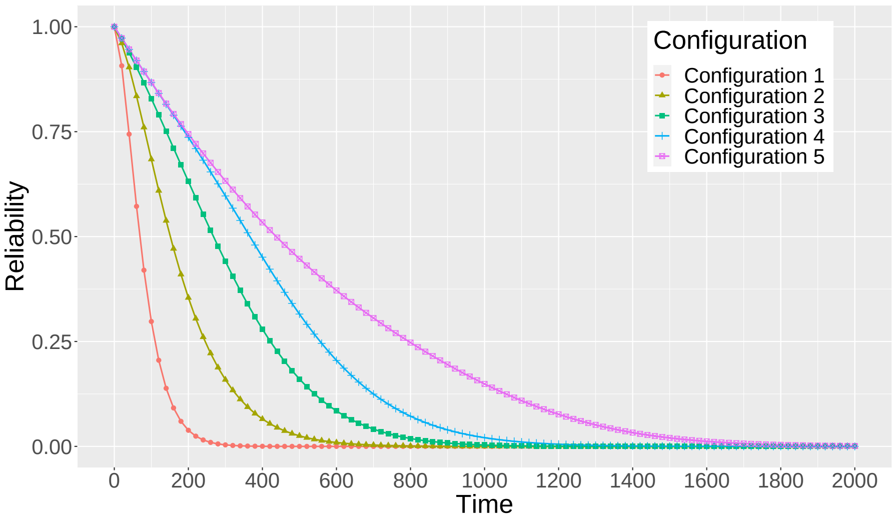


Fig. 14: System MTTF.

7.4 Discussions and Insights

Complexity of System Architecture In this paper, we proposed a fault tree model with a Markov chain for an architecture of a smart city to help system administrators plan complex computing architectures. The model presented in Figure 3 is a high-level representation of the system and is intended to provide an overview of the different components and their interactions. It is not intended to replace existing methodologies in the literature but rather to provide a different perspective on evaluating the dependability of smart city systems. Our goal in this paper was to provide a broad and adaptable model to represent a smart city with three macro-contexts, including smart building, VANET, and smart surveillance. It is acknowledged that the model may appear over-simplistic and may not consider certain aspects of communications redundancy, such as multiple routes to reach the sink node/server. However, it is important to note that the proposed model is not meant to be a detailed and specific model of a single component or system, but rather a high-level representation of the overall smart city architecture. The proposed model uses hierarchical modeling, sequential sensitivity analysis, and KooN gates. The sequential sensitivity analysis method allows for identifying the most important components of the system in terms of their impact on dependability metrics, which can be useful in making decisions regarding system design and configuration. The modeling approach used in this paper is static and manual, which may be considered a limitation. However, it is believed that this approach is suitable for the scope of this work, which focuses on providing a high-level overview of the system and its components.

Detailed Level of Quantification The results presented in the paper are based on a set of assumptions and simplifications made to the system architecture, such as the power grid and cloud infrastructure, in order to make the model more manageable and computationally feasible. It is acknowledged that these simplifications may have an impact on the results. In addition, an exponential distribution was used to model the components' failure rates, which is a common choice in dependability analysis. However, it is acknowledged that different types of distributions, such as Weibull or lognormal, may be more appropriate for certain types of components. Further research will investigate the impact of using different distribution types on the results and to provide a more realistic baseline scenario. Our proposal advances the state-of-the-art by evaluating the dependability of a smart city architecture from a broad perspective, and by combining the use of fault tree models and Markov chains, with a hierarchical modeling strategy and the use of KooN gates, as well as a new method called sequential sensitivity analysis. The proposal presents a novel approach for assessing the availability and reliability of a smart city system by combining the use of fault tree models and Markov chains, with a hierarchical modeling strategy and the use of KooN gates. Furthermore, the sequential sensitivity analysis method allows for identifying the most important components of the system in terms of their impact on dependability metrics. This is the first paper that addresses this strategy.

In this study, a Hierarchical Fault Tree (HFT) analysis was utilized to model the dependability of a smart city system. The HFT approach models the system as a tree-like structure, with the top-level representing the overall system failure and lower levels representing the failures of individual components. Fault Tree Analysis (FTA) is a method for identifying and analyzing the combinations of basic events that can lead to system failure. The mathematical representation of a fault tree is a Boolean function, and the probability of the top event can be calculated using the probability of basic events and the logical operators connecting them. Additionally, this study utilized a new method called sequential sensitivity analysis (SSA) to optimize the metrics evaluated by the proposed models by identifying the system's most important components. Our proposed model is unique in its approach by combining the use of fault tree models and Markov chains, a hierarchical modeling strategy, and the use of KooN gates. In future work, we plan to investigate the impact of different distribution types on the results, provide a more realistic baseline scenario, conduct a performance analysis, and include additional components such as onboard units of vehicles in VANETs to further improve the realism of our model.

8 Conclusion

In this paper, we proposed a novel approach for assessing the availability and reliability of a smart city system, utilizing a combination of a fault tree model and a Markov chain. By investigating three inter-correlated macro-architectures, including smart building, VANET, and smart surveillance, we were able to identify key factors that influence the total dependability of the system. By applying the KooN paradigm and varying the failure requirements of all components, we were able to conduct a sensitivity analysis to identify the most impactful component of the entire system. Our results indicated that the Road Side Unit (RSU) of the VANET macro-component had the greatest impact on system dependability. By varying the number of available RSUs, we were able to demonstrate that even small changes in configuration could result in a significant reduction in downtime. This is particularly important in the case of vehicle traffic control, where even one hour of downtime can be catastrophic. Furthermore, by adopting the sequential sensitivity analysis, we were able to significantly improve the system's Mean Time to Failure (MTTF) from 80.72 hours to 505.29 hours. The case studies presented in this paper provide a practical guide for system administrators to apply our model and assess various configurations for a smart city architecture. Future work includes conducting a performance analysis to evaluate the impact of component availability on response time and throughput of the system. Additionally, other components such as onboard units of vehicles in VANETs can be considered to further improve the realism of the model.

9 Declarations

Ethical Approval Not applicable

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Authors' contributions F.A.S. has managed the process as research leader. F.A.S. and I.F. executed the research and wrote the paper. F.S. and T.A.N. reviewed the manuscript.

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